

ALYSSA L

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[LinkedIn](#) | [GitHub](#) | [AGENTRUN Paper](#)

RESEARCH PROFILE

AI/ML researcher specializing in **failure-mode analysis** and **system-level reliability** for agentic and vision AI. My research philosophy is defined by exposing hidden vulnerabilities—discovering a preprocessing domain-shift in deployed agricultural AI that caused **100% misclassification**, and diagnosing a QLoRA intervention confounder that regressed agent performance from **95.7% to 78.3%** before recovering it to **100%**. I build evaluation frameworks that turn system failures into actionable robustness insights.

Research Interests: AI Safety & Evaluation | Agentic AI Reliability | MCP/Protocol-Level Agentic Systems | Failure-Driven Evaluation Frameworks | Multimodal AI | AI for Biomedicine

EDUCATION

Siddaganga Institute of Technology (SIT) Tumakuru, India
B.E. Information Science & Engineering | **CGPA: 8.66/10** Expected 2028

Relevant Coursework: Advanced Linear Algebra, Probability & Statistics, Optimization Methods, Numerical Methods, Data Structures & Algorithms

PUBLICATIONS & PREPRINTS

AGENTRUN: Failure-Driven QLoRA Specialization for Multi-Tool Agent Behavior in a 7B Language Model 2026
First Author — Published Preprint [DOI: 10.5281/zenodo.22346935](https://doi.org/10.5281/zenodo.22346935)

Controlled QLoRA fine-tuning study exposing a **95.7% → 78.3% regression** and trace-level recovery to **100%**; contributes a failure taxonomy and analysis of category-level train/eval leakage.

CropScout AIoT: A CNN-Based Precision Agriculture Platform 2026
First Author — Preprint in Progress (Expected Oct 2026)

CNN-based plant disease detection (38 classes, 96.56% accuracy) with a documented **preprocessing domain-shift causing 100% misclassification** in edge deployments.

Pawsight: An EfficientNet-Based Diagnostic System for Canine Health 2026
First Author — Preprint in Progress (Expected Oct 2026)

EfficientNetB0-based veterinary screening (7 conditions, 98.5% accuracy) with Grad-CAM explainability.

Smart Vision Stick: An AI-Driven Object Detection and Guidance System 2026
Co-Author (5th) Under Review at ICEI 2026

YOLOv8-based assistive system for visually impaired individuals; 93.5% object detection accuracy.

RESEARCH EXPERIENCE

Agentic AI Systems — Controlled QLoRA Specialization Study 2025–Present
Independent Research — Siddaganga Institute of Technology

- Designed a controlled intervention study to test whether small-sample QLoRA fine-tuning reliably improves multi-tool agent execution—found it **regressed from 95.7% to 78.3%**.
- Performed trace-level failure diagnosis across 23 held-out tasks, identifying two root causes: a serving-layer parser defect and training-coverage gaps in chained/error-handling patterns.
- Formally documented the recovery to **100%** while transparently analyzing the category-level train/eval leakage introduced by the failure-driven correction—contributing a methodological template for agent evaluation.
- Established MCP-based execution traces to enable **auditable, governance-ready agent loops**.

Computational Research — GROMACS / Corneal Systems 2026–Present
Independent Research — Siddaganga Institute of Technology

- Conducting molecular-dynamics simulations using **GROMACS** for corneal biomolecular systems: force fields, topology, solvation, minimization, equilibration, and production MD.
- Developing **HPC/GPU research workflows** with job scripting and array management on cluster systems.
- Exploring AI/ML regression models to predict donor corneal endothelial recovery post-hypothermia—bridging MD trajectory data with predictive ML (interdisciplinary with ECE and Chemical Engineering).

SELECTED AI / ML PROJECTS

AGENTRUN — Agentic AI Runtime Python, QLoRA, MCP, Qwen2.5-7B
github.com/alyssa2ai/AGENTRUN

- Built a minimal agent runtime from scratch: **think-act-observe** loop, tool execution, short-term memory, and structured trace logging.
- Implemented QLoRA training pipeline on a free-tier T4 GPU; integrated MCP client/server with JSON-RPC tool exe-

cution.

- Full reproducibility: training data, scripts, benchmark, and result JSONs released open-source.

Waggle-mcp — Persistent Graph-Backed Memory for Agents

Python, MCP, SQLite, Neo4j

github.com/alyssa2ai/Waggle-mcp

- Built an MCP server with persistent graph-backed memory, hybrid retrieval, recursive context assembly, and provenance tracking.
- Implemented contradiction handling and Graph Studio visualization for agent decision traces.
- Included CI/CD, testing, and full documentation.

NanoGPT-from-Scratch — Transformer Architecture

PyTorch, Transformers, Deep Learning

github.com/alyssa2ai/nanogpt-from-scratch

- Implemented a decoder-only Transformer from scratch: causal masking, multi-head self-attention, positional embeddings, autoregressive generation.
- Trained character-level language models on Tiny Shakespeare; analyzed training dynamics across configurations.

Agentic RAG Researcher — Autonomous Research Agent

LangChain, LangGraph, ChromaDB, Groq Llama 3.3

github.com/alyssa2ai/agentic-rag-researcher

- Built an autonomous research workflow: recursive reasoning, persistent vector memory, multi-tool retrieval, iterative critique/refinement.
- **In progress:** Extending to multimodal retrieval and multi-agent orchestration.

AWARDS & ACHIEVEMENTS

- **1st Place — KIT SPOC Innovathon 2026**
- **2nd Place — Scaler ProdX Hackathon:** Built an Agentic AI B2B workflow automation platform.
- **1st Place — Technesium 2025 Line Following Robot Competition**
- **Top 5 — RVIT Hackathon**
- **Top 10 — Smart India Hackathon Internal Competition**

LEADERSHIP & COMMUNITY IMPACT

Core Team Member — CORSIT, SIT

2025–Present

Computer Society of India Student Chapter

Tumakuru, India

- **Scaled interdisciplinary events** managing logistics for **1,300+ participants** across 9 engineering disciplines (ROBO-COR'26).
- Mentored **37+ teams (148+ participants)** in hands-on robotics/AI development.

Hackathon Organizer & Mentor — AI Brewery, SIT

2025–Present

AI/ML Student Community

Tumakuru, India

- Organized AI-focused hackathons; mentored student teams through AI/ML and robotics project development.
- Supported students in practical AI workflows and technical problem-solving.

Secretary — LEO District 317A

2026–Present

Leo Clubs International

- District-level youth leadership role supporting community-service and social-impact initiatives.
- Contributed to health-awareness and community outreach activities.

TECHNICAL SKILLS

Programming: Python, C++, Java, JavaScript, TypeScript, SQL

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenCV, MediaPipe, LLMs, RAG, Generative AI, LangChain, LangGraph, Grad-CAM, QLoRA, MCP

Systems & Tools: GROMACS, CUDA, SLURM, ChromaDB, PostgreSQL, MongoDB, Docker, Git, FastAPI

HPC & Scientific Computing: Molecular Dynamics Simulations, Force Fields, Energy Minimization, HPC/GPU Workflows

Research Methodology: Controlled Experiment Design, Failure-Mode Analysis, Ablation Studies, Statistical Model Evaluation, Trace-Level Debugging for Agents